...\Ionosph_collis_profile_test_2_27_2015.tex

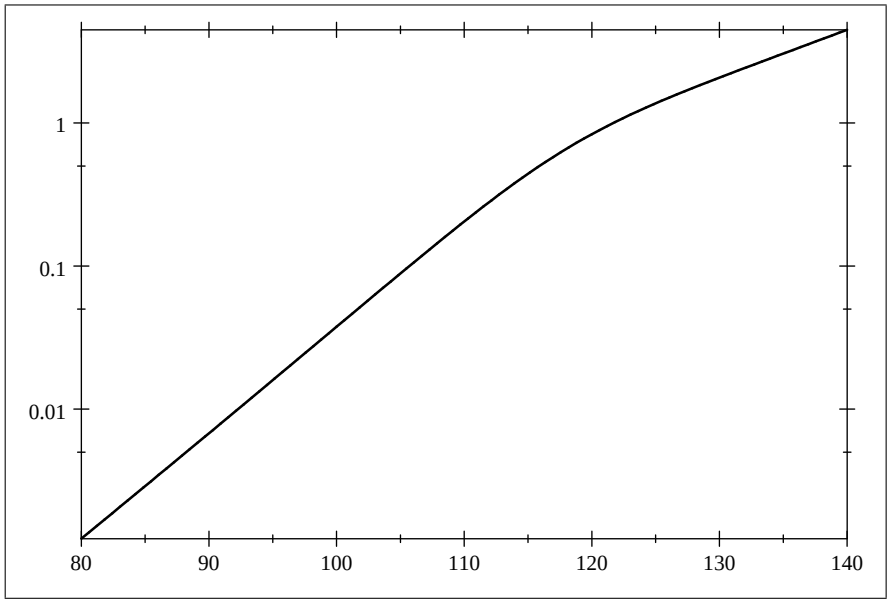
October 15, 2026

1 Definitions:

$$\psi(h) = \frac{1}{\kappa_e(h)\kappa_i(h)} = \frac{\exp\left(-\frac{h-94.5}{3}\right)}{1+\exp\left(\frac{h-120}{15}\right)} + \frac{\exp\left(-\frac{h-71.7}{6}\right)}{1+\exp\left(-\frac{h-120}{15}\right)}$$

$$\Theta_0 = \sqrt{\frac{m_e\nu_{en}}{m_i\nu_{in}}} = 1.348\,399\,725 \times 10^{-2}$$

$$\kappa_e(h) = \frac{\Omega_e}{\nu_{en}} = \frac{1}{\Theta_0\sqrt{\psi(h)}}, \qquad \kappa_i(h) = \frac{\Omega_i}{\nu_{in}} = \frac{\Theta_0}{\sqrt{\psi(h)}}$$



$\kappa_i(h)$